

windowfunctions

March 1, 2026

PySpark Window Functions – One Table Cheat Sheet

Category	Function	Syntax (PySpark DataFrame)	Description
Window Spec	Define Window	<code>from pyspark.sql.window import Windoww = Window.partitionBy("dept").orderBy("salary")</code>	Creates window specification
	Partition Only	<code>Window.partitionBy("dept")</code>	Divides data into groups
	Order Only	<code>Window.orderBy("salary")</code>	Defines row ordering
	Partition + Order	<code>Window.partitionBy("dept").orderBy("salary")</code>	Most common window definition
	Rows Frame	<code>w.rowsBetween(-1,1)</code>	Row-based sliding window
	Range Frame	<code>w.rangeBetween(-1000,1000)</code>	Value-based window

Ranking Functions

Function	Syntax	Description
<code>row_number()</code>	<code>from pyspark.sql.functions import row_numberdf.withColumn("rn", row_number().over(w))</code>	Unique sequential number
<code>rank()</code>	<code>from pyspark.sql.functions import rankdf.withColumn("rank", rank().over(w))</code>	Same rank for ties, gaps allowed
<code>dense_rank()</code>	<code>from pyspark.sql.functions import dense_rankdf.withColumn("drank", dense_rank().over(w))</code>	Same rank, no gaps
<code>percent_rank()</code>	<code>from pyspark.sql.functions import percent_rankdf.withColumn("pr", percent_rank().over(w))</code>	Relative rank (0–1)
<code>ntile(n)</code>	<code>from pyspark.sql.functions import ntiledf.withColumn("bucket", ntile(4).over(w))</code>	Divides rows into n buckets

Analytical (Value) Functions

Function	Syntax	Description
lag()	<code>from pyspark.sql.functions import lagdf.withColumn("prev_sal", lag("salary",1).over(w))</code>	Previous row value
lead()	<code>from pyspark.sql.functions import leaddf.withColumn("next_sal", lead("salary",1).over(w))</code>	Next row value
first()	<code>from pyspark.sql.functions import firstdf.withColumn("first_sal", first("salary").over(w))</code>	First value in window
last()	<code>from pyspark.sql.functions import lastdf.withColumn("last_sal", last("salary").over(w))</code>	Last value in window
nth_value()	<code>from pyspark.sql.functions import nth_valuedf.withColumn("nth", nth_value("salary",2).over(w))</code>	Nth row value
cume_dist()	<code>from pyspark.sql.functions import cume_distdf.withColumn("cd", cume_dist().over(w))</code>	Cumulative distribution

Aggregate Window Functions

Function	Syntax	Description
sum()	<code>from pyspark.sql.functions import sumdf.withColumn("sum_sal", sum("salary").over(w))</code>	Running sum
avg()	<code>from pyspark.sql.functions import avgdf.withColumn("avg_sal", avg("salary").over(w))</code>	Running average
min()	<code>from pyspark.sql.functions import mindf.withColumn("min_sal", min("salary").over(w))</code>	Running minimum
max()	<code>from pyspark.sql.functions import maxdf.withColumn("max_sal", max("salary").over(w))</code>	Running maximum
count()	<code>from pyspark.sql.functions import countdf.withColumn("cnt", count("salary").over(w))</code>	Running count
collect_list()	<code>from pyspark.sql.functions import collect_listdf.withColumn("list", collect_list("salary").over(w))</code>	Collects values as list
collect_set()	<code>from pyspark.sql.functions import collect_setdf.withColumn("set", collect_set("salary").over(w))</code>	Collects unique values

Common Exam Patterns

Use Case	Window Spec
Running total	<code>Window.partitionBy("dept").orderBy("date").rowsBetween(Window.unboundedPreceding, Window.currentRow)</code>
Top N per group	Use <code>row_number()</code> + filter
Remove duplicates (latest record)	<code>row_number().over(partitionBy("id").orderBy(desc("date")))</code>

Use Case	Window Spec
Compare current vs previous	<code>lag()</code>
Salary difference	<code>col("salary") - lag("salary",1).over(w)</code>